

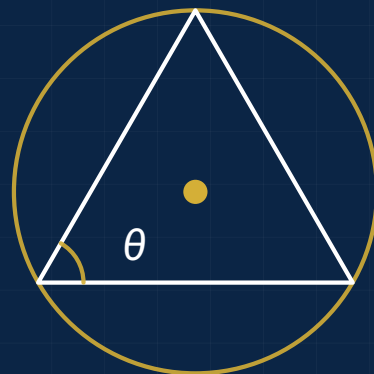


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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 12

3D Pythagoras & Trigonometry

Every 3D question hides a flat right-angled triangle. Find it, draw it on its own, and use Pythagoras or SOH CAH TOA — often twice.



3

VARIANTS

15

PAST-PAPER PARTS

25

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Lengths in 3D

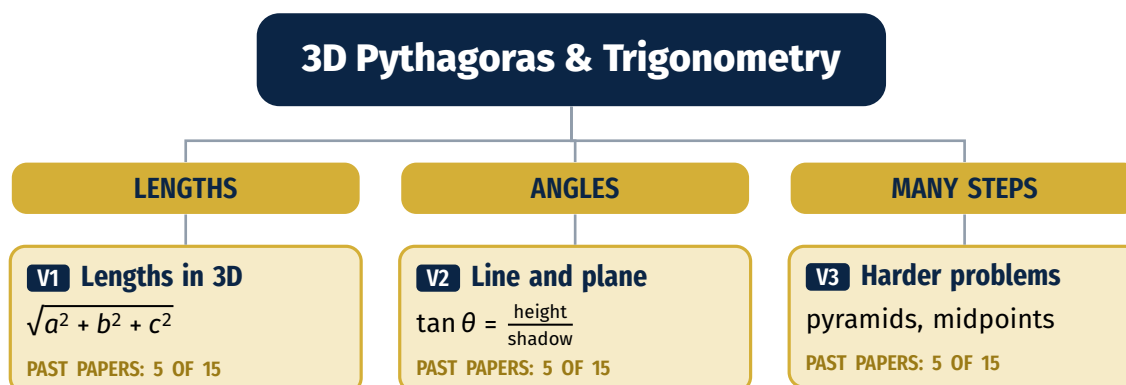
V2 Angle between a line and a plane

V3 Harder 3D problems

TOPIC MAP • EVERY VARIANT ON ONE PAGE

3D trigonometry at a glance



Your notes


THE KEY HABIT

DRAW THE FLAT TRIANGLE

Take the triangle out of the 3D picture and draw it flat.

★ How often does each variant come up?

13 past-paper questions (Jan 2020 → Nov 2025) hold 15 parts; 25 practice parts — grouped by type.

	PAST PAPERS (OUT OF 15)	PRACTICE PARTS
V1 Lengths in 3D	5	9
V2 Angle: line and plane	5	10
V3 Harder 3D problems	5	6

WHAT THIS TELLS YOU One question on most papers, near the end (Q17–26, 3–6 marks). It is two right-angled triangles in a row: the base diagonal first, then the height.

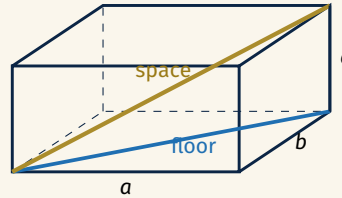
BEFORE YOU START

The big idea

Floor first, then up.

To reach a far top corner: first cross the **floor** (Pythagoras on the base), then go **up** (Pythagoras again).

Shortcut: space diagonal
 $= \sqrt{a^2 + b^2 + c^2}$.



Your notes

**Angle between a line and a plane****The line**

AH

The slanted line in the question.

Its shadow

FH

Drop the top end straight down to the plane.

The angle

$$\tan \theta = \frac{\text{height}}{\text{shadow}}$$

Between the line and its shadow.

**Useful 3D facts****Square base side s**

diagonal $s\sqrt{2}$

Half of it reaches the centre.

Cube side s

space diagonal $s\sqrt{3}$

$\sqrt{3}s^2$.

Vertical line

90° to the base

Gives the right angle.

**CALCULATOR FIRST, THEN CHECK**

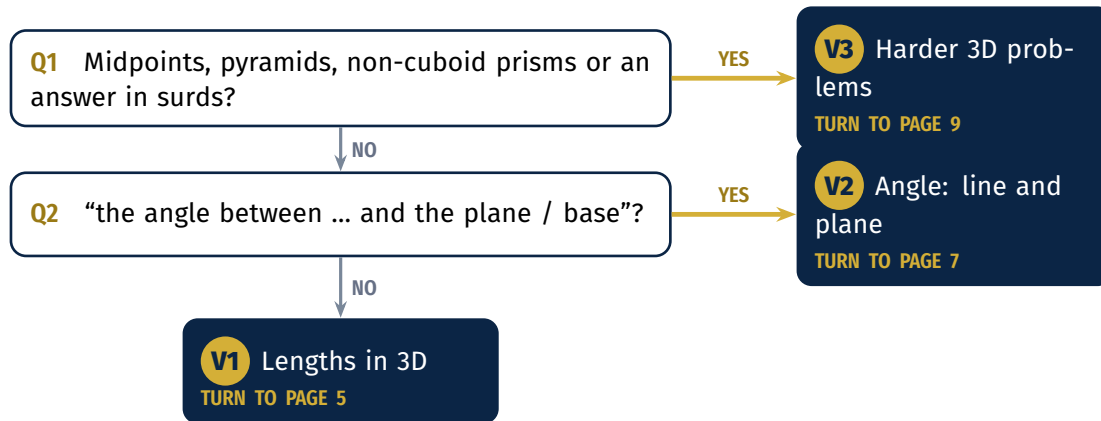
Keep the floor diagonal as $\sqrt{106}$ (or on Ans) — don't round before the second step.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?


 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

Lengths in 3D

Base diagonal, then the vertical height; or
 $\sqrt{a^2 + b^2 + c^2}$ in a cuboid.

SEEN IN PAST PAPERS

5 / 15

PRACTICE QUESTIONS: 9


 Your notes

Came up in: Jun 2022 P1HR Q18 • Jun 2025 P1HR Q20 • Nov 2025 P1H Q26 • Jun 2023 P2HR Q22(a) • Oct/Nov 2020 P1HR Q17(b)

? What is it?**In plain words**

A line through the inside of a solid is the hypotenuse of a right triangle standing on the **floor**.

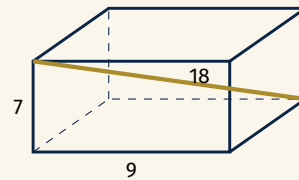
Cuboid: $d = \sqrt{a^2 + b^2 + c^2}$.

Backwards: d known, find a side by subtracting squares.

IT LOOKS LIKE

$$\sqrt{54}$$

$$FC = 18 \text{ find } BC$$

**How do I do it?****1 Ends**

The two points of the line.

2 Floor

Right triangle on the base.

3 Up

Right triangle with the height.

4 OrOne step: $\sqrt{a^2 + b^2 + c^2}$.**FLOOR, THEN UP**

Or all three squares at once.

V1 Lengths in 3D

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“the distance ... in a straight line through the cube”

→ $\sqrt{a^2 + b^2 + c^2}$

A diagonal given, a side wanted

→ **Subtract squares**“in the form $\sqrt{a}$ ”→ **Leave it as $\sqrt{54}$** **WORKED EXAMPLE** **PAST PAPER**

June 2025 • Paper 1HR • Q20 • 3 marks

Cuboid $ABCDEFGH$: $AB = 9$ cm, $AF = 7$ cm and the space diagonal $FC = 18$ cm. Calculate BC (3 s.f.).

STEP 1 $FC^2 = AB^2 + AF^2 + BC^2$ ◀ three edges

STEP 2 $324 = 81 + 49 + BC^2$ ◀ substitute

STEP 3 $BC^2 = 194$ ◀ subtract

STEP 4 $BC = 13.93 \dots$ ◀ root

ANSWER 13.9 cm

✓ shorter than 18 ✓

**WATCH OUT**× $BC = 18 - 9 - 7$ ✓ **subtract the squares**

Pythagoras works with squares, not lengths.

**EXAM TIP**

Sketch the flat triangle with its sides labelled.

**Check yourself**

calculator first • answers on page 11

1.1

Jun 2022 P1HR

A cube of side 6 cm; T is the midpoint of AB , V of CH . The steps from T to V are 3, 6, 3 cm. Find TV .

Answer:

1.2

Oct/Nov 2020 P1HR

Prism on a 12 cm square base; B is 19 cm above G ; E is the opposite base corner. Find BE .

Answer:

1.3

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A cuboid is 3 cm by 4 cm by 12 cm. Length of its space diagonal?

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Angle between a Line and a Plane

Drop the line onto the plane (its shadow);

$$\tan \theta = \frac{\text{height}}{\text{shadow}}$$

SEEN IN PAST PAPERS

5 / 15

PRACTICE QUESTIONS: 10



Your notes

Came up in: Jan 2020 P1H Q19 • Oct/Nov 2020 P1HR Q17(a) • May 2021 P1H Q22 • Jan 2022 P2HR Q19 • Jun 2025 P2H Q24

? What is it?

In plain words

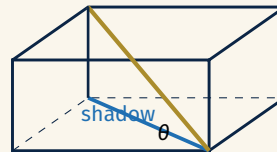
Shine a light straight down: the line casts a **shadow** on the plane. The angle is between the line and its **shadow**.

The **height** and the **shadow** make a right angle — use tan.

IT LOOKS LIKE

AH and plane EFGH

EC and ABCD



How do I do it?

1 Line

Name its two ends.

2 Shadow

Drop the top end vertically; join to the bottom end.

3 Shadow length

Usually a base diagonal (Pythagoras).

4 Angle

$$\tan^{-1} \frac{\text{height}}{\text{shadow}}$$

LINE, SHADOW, TAN

The angle sits at the bottom end.

V2 Angle between a Line and a Plane

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“the angle between AH and the plane $EFGH$ ” → **Shadow FH ; height AF**

“a vertical pole AE at the corner A ” → **Shadow AC on the field**

“Give your answer correct to 3 s.f.” → **Keep the shadow length exact**

WORKED EXAMPLE **PAST PAPER**

January 2020 · Paper 1H · Q19 · 4 marks

Cuboid: $EH = 9$ cm, $HG = 5$ cm, $GB = 6$ cm (height). A is above F . Find the angle between AH and the plane $EFGH$ (3 s.f.).

STEP 1 shadow of $AH = FH$ ◀ A is above F **STEP 2** $FH = \sqrt{9^2 + 5^2} = \sqrt{106}$

◀ floor diagonal

STEP 3 $\tan \theta = \frac{6}{\sqrt{106}}$

◀ height over shadow

STEP 4 $\theta = 30.20 \dots$

◀ inverse

ANSWER 30.2°

✓ shallow line, small angle ✓

WATCH OUT

$$\times \tan \theta = \frac{6}{9}$$

$$\checkmark \frac{6}{\sqrt{106}}$$

The shadow runs across the floor diagonally, not along an edge.

EXAM TIP

Draw the flat triangle: height, shadow, line — with the right angle marked.

Check yourself

calculator first · answers on page 11

2.1

Jan 2022 P2HR

Field $ABCD$: $AB = 18$ m, $AD = 8$ m; vertical pole $AE = 12$ m. Angle between EC and $ABCD$?

Answer:

2.2

Oct/Nov 2020 P1HR

C is 10 cm above H ; F is the opposite corner of a 12 cm square base. Angle between CF and the base?

Answer:

2.3

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A cuboid has base 6 cm by 8 cm and height 5 cm. Angle between a space diagonal and the base?

Answer:

V3**VARIANT 3** MOST COMMON IN THE EXAM**Harder 3D Problems**

Find the right triangle through midpoints or a pyramid's apex; often a length first, then the angle.

SEEN IN PAST PAPERS

5 / 15

PRACTICE QUESTIONS: 6


 Your notes

Came up in: Jan 2021 P2HR Q22 • Jan 2022 P1H Q21 • Jun 2023 P2HR Q22(b) • Jun 2025 P2HR Q23 • May 2024 P1HR Q22

? What is it?**In plain words**

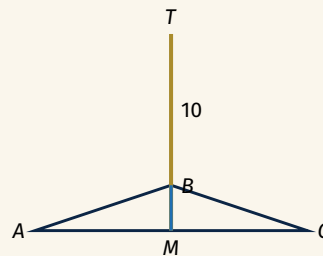
The triangle you need goes through a **midpoint**, the **centre** of a base, or a point **vertically above** another.

Find the missing horizontal length first (trig in the base), then the angle with the vertical height.

IT LOOKS LIKE

T above B

M midpoint of AC

**How do I do it?****1 Special points**

Midpoint, centre, foot of the vertical.

2 Horizontal

Its length in the base (trig / Pythagoras).

3 Vertical

The height above it.

4 Angle

$$\tan^{-1} \frac{\text{vertical}}{\text{horizontal}}$$

BASE FIRST, THEN UP

One flat triangle at a time.

V3 Harder 3D Problems

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"M is the midpoint of AC" → **$BM \perp AC$ in an isosceles base**

"Q is the midpoint of AB ... angle EQM" → **QM = half the side**

"angle between PU and the plane is 30° " → **Height = diagonal $\times \tan 30^\circ$**

WORKED EXAMPLE **PAST PAPER**

January 2021 • Paper 2HR • Q22 • 4 marks

ABC is horizontal with $AB = BC = 8$ cm and angle $ABC = 140^\circ$. T is 10 cm vertically above B; M is the midpoint of AC. Find the angle between MT and the plane ABC (1 d.p.).

STEP 1 angle $ABM = 70^\circ$ ◀ BM bisects the isosceles apex

STEP 2 $BM = 8 \cos 70^\circ = 2.736$ ◀ base triangle

STEP 3 $\tan \theta = \frac{10}{2.736}$ ◀ height over shadow

STEP 4 $\theta = 74.69 \dots$ ◀ inverse

ANSWER 74.7°

✓ T high above a near point: steep ✓

WATCH OUT✗ $BM = 8 \sin 70^\circ$ ✓ $8 \cos 70^\circ$ BM is adjacent to the 70° angle.**EXAM TIP**

Redraw the base triangle flat, then the vertical triangle beside it.

Check yourself

calculator first • answers on page 11

3.1

Jun 2025 P2HR

Prism on rectangle ABCD: $AC = 40$, angle $CAE = 35^\circ$ (E above C); M midpoint of AC. Angle CME?

Answer:

3.2

Jun 2023 P2HR

Cuboid: $PR = 42$ cm, angle between PU and PQRS is 30° (U above R); M midpoint of PR. Angle UMR?

Answer:

3.3

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Square-based pyramid: base 10 cm, vertical height 12 cm. Angle between a sloping edge and the base?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked in Python (check_answers.py).

1.1 $\sqrt{54}$ cm

1.2 25.5 cm

1.3 13 cm

2.1 31.4°

2.2 30.5°

2.3 26.6°

3.1 54.5°

3.2 49.1°

3.3 59.5°



Your notes

THE WHOLE TOPIC ON ONE PAGE

3D cheat sheet



Your notes
V1 Lengths 5/15

$$\sqrt{a^2 + b^2 + c^2}$$

Floor, then up.

V2 Line & plane 5/15

$$\tan \theta = \frac{6}{\sqrt{106}}$$

Line, shadow, tan.

V3 Harder 5/15

$$8 \cos 70^\circ$$

Base first, then up.

**TOP 3 LOST MARKS**

1. an edge used instead of the floor diagonal ×
2. angle at the wrong end ×
3. rounding the first length ×

BOTTOM LINE Find the flat triangle · floor, then up · line, shadow, tan · keep exact values.